



Pharm*Achieve*

COMPOUNDING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ABHR Alcohol-Based Hand Rub

API Active Pharmaceutical Ingredient

BSC Biological Safety Cabinet

BUD Beyond-Use Date

CDF Compromised Dosage Form

C-PEC Containment Primary Engineering Control

HEPA High-Efficiency Particulate Air

IODF Intact Oral Dosage Form

IPA Isopropyl Alcohol

IV Intravenous

MFR Master Formulation Record

NAPRA National Association of Pharmacy Regulatory Authorities

PEC Primary Engineering Control

PO Per Os (Oral)

PPE Personal Protective Equipment

SDS Safety Data Sheet

SOP Standard Operating Procedure

BACKGROUND

Health Canada defines compounding as the **mixing or combining of ≥ 2 ingredients** (with ≥ 1 being a drug or pharmacologically active component) to create a final product

Compounding should only be done if there is a **therapeutic need or lack of product availability** (e.g. shortage of commercially available product)

A **patient-healthcare professional relationship** must exist in pharmacy compounding settings

Includes the preparation, mixing, assembling, packaging and labelling of safe and effective medicines for immediate use*

Medicines contain preservatives, suspending agents, surfactants, water, flavouring and colouring agents

* A pharmacy may compound or repackage multiple units, not for immediate use, in very limited quantities during a single process, in anticipation of a prescription

SKILLS REQUIRED FOR COMPOUNDING

Accurate calculations of quantities of the active pharmaceutical ingredient(s) (API) and excipients

Know the relevant federal, provincial, territorial legislation and regulations for compounding, including guidelines

Good laboratory practices

Understand physical and chemical properties

Proper documentation

Perform risk assessments

Prevent compounding errors

Take corrective steps when irregularities occur

Aseptic technique for sterile compounding

Quality assurance

Label printing and packaging in accordance with the pharmaceutical company standards

Correct operation of equipment, materials, and automated instruments

COMPOUNDING GUIDELINES AND STANDARDS

- The National Association of Pharmacy Regulatory Authorities (NAPRA) has Model Standards for:
 - Non-sterile compounding
 - Non-hazardous sterile compounding
 - Hazardous sterile compounding
- Model Standards include information on:
 - Compounding facility (space, equipment, personal protective equipment (PPE))
 - Personnel (roles, training, competency)
 - Risk assessments
 - Quality assurance
 - Compliance
 - Competencies

TYPES OF COMPOUNDING

Non-sterile compounding occurs when medications are prepared in a non-sterile environment whilst maintaining the product's integrity

Examples:

- Topical creams
- Topical ointments
- Oral medications

TYPES OF COMPOUNDING

Sterile compounding occurs when medications are prepared in an environment that prevents contamination

Examples:

- Nasal inhalation solutions
- Respiratory therapy solutions
- Injections (e.g. intramuscular, intravenous, intrathecal, intradermal, subcutaneous)
- Irrigation solutions for wounds and body cavities (e.g. thoracic, spinal, abdominal, pelvic)
- Ophthalmic drops and ointments
- Otic drops for intratympanic administration
- Parenteral nutrition
- Dialysis solutions
- Allergen extracts
- Topical preparations where sterility is essential (i.e. burn patients)
- Radiopharmaceuticals

A top-down view of a medical workspace on a light blue surface. It includes a silver laptop with a medical website on its screen, a silver stethoscope, a spiral-bound notebook with a black pen, a pair of glasses, a black smartphone, and a white coffee cup with a lid. A large, white, rounded rectangular box with a red border is centered over the image, containing the text 'NON-STERILE COMPOUNDING'. A diagonal watermark reading 'Licensed to Shreyash Salunkhe' is visible across the image.

NON-STERILE COMPOUNDING

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Gloves and gown should be worn for compounding of most substances
- Face masks are optional unless the substance is hazardous
- Hair, beard and shoe covers should be worn when compounding hazardous preparations
- Eye protection should be worn if there is a risk of splashing or powder exposure

LIQUID VEHICLES

Oral

- Properties: emulsify, suspend and dissolve
- Use: dilute drugs to a suitable volume for proper dosing, adherence and stability

Water

- Most common formulation of a solution
- Purified by sterilization
- Injections: pyrogen-free distilled water

COMPOUNDING TECHNIQUES

Eutectic

- Solid + Solid = Liquid

Levigation

- Powder + Liquid = Liquid or Thin Paste

Trituration

- Grinding of powder(s) with a mortar and pestle

Geometric Dilution

- Used when mixing a small amount of a drug with a large amount of another ingredient or diluent
- The drug present in smaller quantity is placed in the mortar with an equal amount of the other ingredient then the two materials are triturated until they are well mixed

EXAMPLE OF COMPOUNDING PROCEDURE

- 1 Calculate and measure the correct quantity of each ingredient
- 2 Place the powder product in a glass mortar and triturate until fine powder is achieved
- 3 Add co-solvents, e.g. isopropyl alcohol (IPA), to the fine powder to form a smooth paste
- 4 Use shear force mixing or milling to create a homogenous mixture
- 5 Package and label the product, counsel on safety concerns and beyond-use date (BUD)

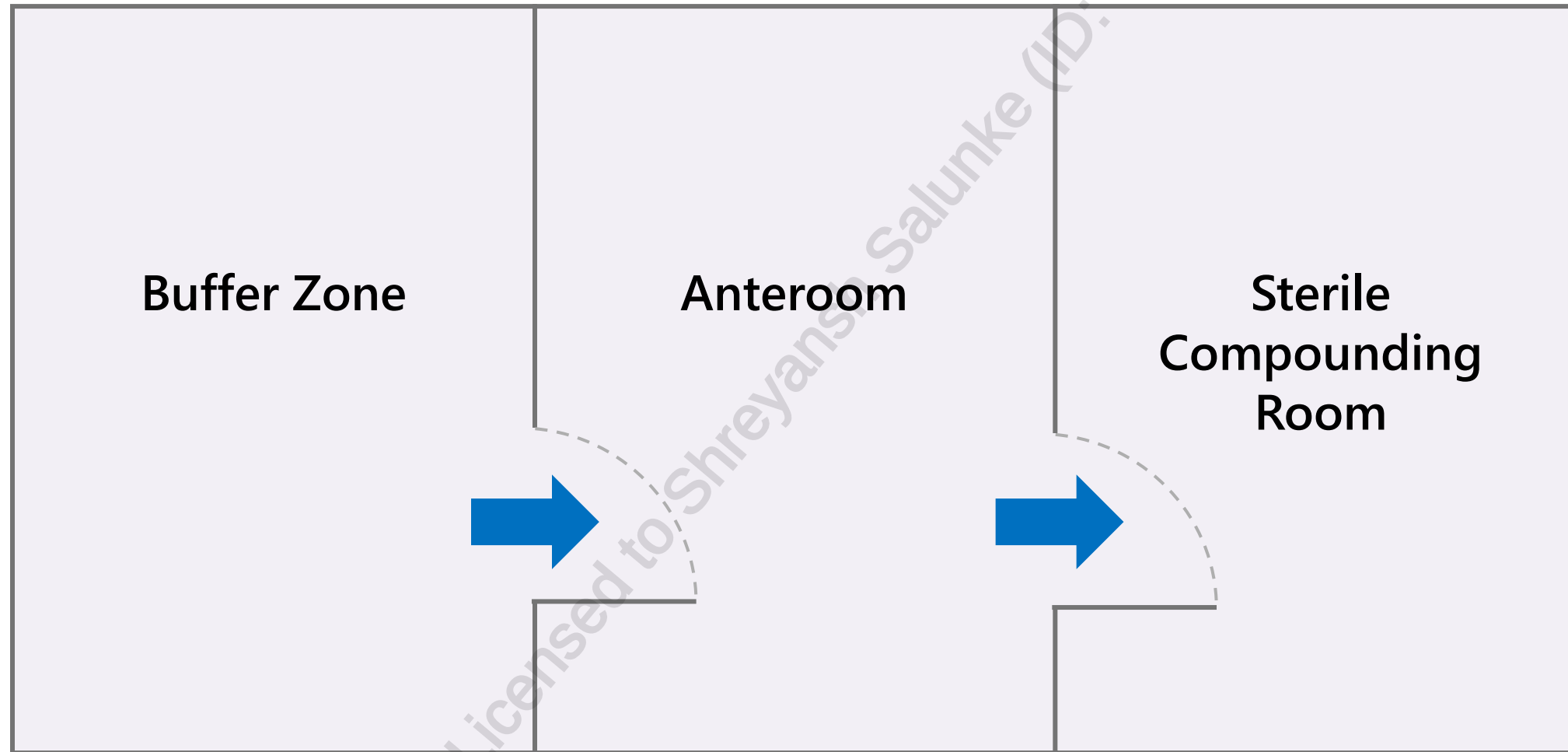
SAFETY PRECAUTIONS

- Clean spillages right away to reduce harm
- Protect from dust, pollen, mould, bacteria, smoke and asbestos
- Use dust-resistant safety goggles
- Disposable gowns
- Proper safety gloves
- Proper safety shoes
- Filters must be certified
- Laminar air flow hoods – see hospital distribution lecture for further information

A top-down view of a medical workspace on a light blue surface. It includes a silver laptop with a medical website on its screen, a black stethoscope, a spiral-bound notebook with a black pen, a pair of glasses, a black smartphone, and a white coffee cup with a lid. A large, white, rounded rectangular box with a red border is centered over the laptop.

STERILE COMPOUNDING

STERILE COMPOUNDING - FACILITY DESIGN



STERILE ROOM: PRIMARY ENGINEERING CONTROL

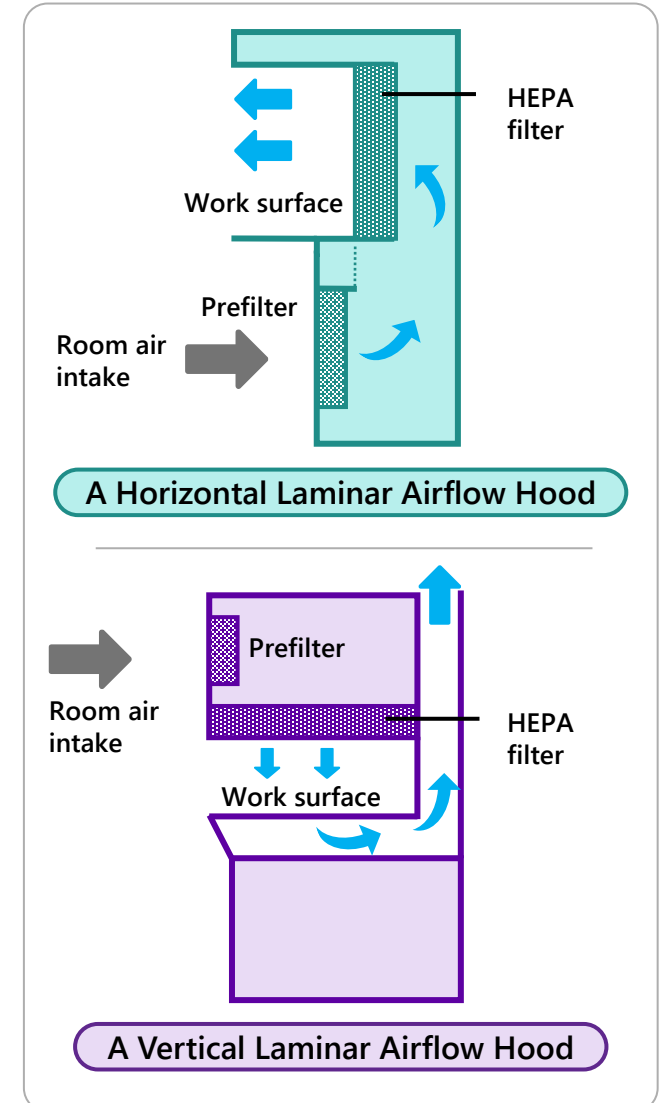
- The primary engineering control (PEC) ensures a safe air quality environment, providing unidirectional (laminar) air flow in the work area
- High-Efficiency Particulate Air (HEPA) filter removes nearly all bacteria from air and prevents entry of contaminated room air into hood
- A containment primary engineering control (C-PEC) is used to compound hazardous sterile preparations e.g. biological safety cabinet (BSC) Class II
- The sterile compounding supervisor ensures certification is complete
- Clean interior with 70% IPA (or equivalent); dress appropriately during use
- Turn on 15-30 mins prior to use; work should always be ≥ 15 cm into hood
- Operate continuously during every sterile compounding activity

Horizontal air flow

- Air circulates from the back to the front
- Used for **most sterile IV preparations**
- Order of cleaning: walls and floor should be **back to front**
- Place objects next to each other

Vertical air flow

- Air circulates from top down
- Used for **chemotherapy** and **hazardous agents** since air flows away from the person
- Order of cleaning: walls should be **top down**, floor should be **right to left** on surface



PERSONAL PROTECTIVE EQUIPMENT (PPE)

- Pair of shoe covers or dedicated shoes
- Hair cover
- Beard cover (if required)
- Surgical mask
- Non-shedding gowns enclosed at the neck with sleeves that fit snugly around the wrists
- Pair of non-powdered sterile gloves which cover the cuffs of the gown

STERILE COMPOUNDING PROCEDURES

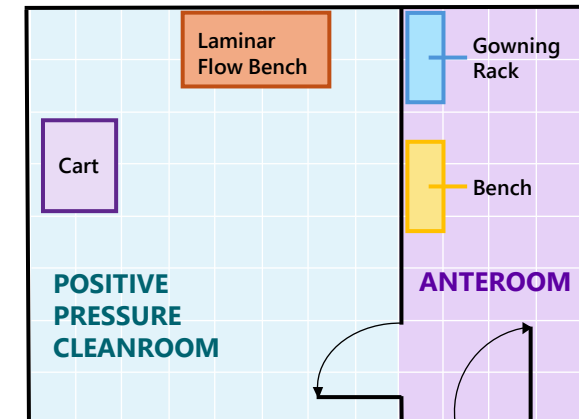
PRIOR TO ENTERING THE ANTEROOM (WHILE IN THE BUFFER ZONE)

- Remove all personal outer garments e.g. jackets, hats, sweaters, outdoor shoes
- Remove jewelry from fingers, wrists, forearms, face, tongue, ears and neck (including personal electronic devices)
- Remove all cosmetics (e.g. makeup, false eyelashes, perfume, hair products, henna and paper tattoos) as these products generate particles that are possible sources of contamination
- Tie up long hair
- Remove nail polish and other nail applications. Ensure nails are trimmed
- Ensure the skin of hands and forearms is undamaged
- Change into low-shedding apparel suitable for a controlled area (e.g. scrubs)
- Wear pants that fully cover the legs, closed shoes, and socks
- Wash hands

STERILE COMPOUNDING PROCEDURES

CONDUCT WHILE IN THE CONTROLLED AREAS (CLEAN ROOM AND ANTEROOM)

- Food items, drinks, chewing gum, candy, and cigarettes (or other smoking products) are prohibited
- All access doors to controlled areas must be kept closed
- Access to controlled areas is restricted to personnel with specific responsibilities in the controlled areas
- All personnel in the controlled areas must follow specified hand hygiene and garbing procedures
- Only essential conversations are allowed, to minimize the risk of particulate contamination. Coughing, sneezing and talking in the direction of the laminar airflow workbench should also be avoided



STERILE COMPOUNDING PROCEDURES

ENTERING THE ANTEROOM (GARBING PROCEDURE)

- While on the “dirty” side of the demarcation line, don a hair net, then beard cover (if required), and then a face mask
- Whilst stepping over the demarcation line to the “clean” side, don shoe covers
- Under warm running water, use a nail pick to remove debris under fingernails, wash hands and forearms (up to elbows) with warm water and soap for at least 30 seconds, dry with a disposable, lint-free towel
- Apply alcohol-based hand rub (ABHR)
- Don a gown* (closed at the neck and elastic cuffs)
- Apply ABHR to hands and allow hands to dry
- Don sterile gloves** and routinely disinfect with sterile 70% IPA (or equivalent) during the compounding process
- Shoe covers or dedicated shoes are required at all times in the clean area of the anteroom and the clean room. Dedicated shoes must not be worn outside of these areas, and must be cleaned and disinfected once a week

*Chemotherapy gown is needed for hazardous products

**Two pairs of gloves are needed for hazardous products

STERILE COMPOUNDING PROCEDURES

WHILE IN THE STERILE ROOM

- Clean the hood with lint-free sterile wipes and 70% IPA
- Allow the hood to run for 30 minutes before use
- Disinfect all supplies with 70% IPA (i.e. ampules, IV bags)
- Place materials at least 6 inches into the hood to avoid air obstruction
- Inspect materials for damage, particles or expiration
- Swab the vial stopper with 70% IPA
- Insert the needle bevel up at a 45-60° angle into the vial
- Inject air equal to the volume of drug being withdrawn into the vial
- Withdraw the medication from the vial
- Withdraw diluent and add to vial if reconstitution is needed, mix gently by swirling NOT shaking
- Swab the vial, IV bag port, etc. that the withdrawn medication is to be injected into
- Inspect the final product for particles or leaks
- Label the product
- Dispose of sharps and hazardous waste in appropriate containers
- Record all lot numbers and expiration dates for the products used



Record cleaning and disinfecting activities in the maintenance log

STERILE COMPOUNDING PROCEDURES

ANTEROOM AFTER LEAVING THE STERILE ROOM (DOFFING PPE)

While on the dirty side of the demarcation line in the anteroom:

- Remove gloves (avoid touching the outer part of the gloves)
- Remove gown by pulling it away from the body and rolling it into a ball
- Remove mask by removing ear straps from the ears
- Remove shoe covers (dedicated shoes remain in the anteroom)
- Hand hygiene

A top-down view of a desk with various items: a laptop with a website on the screen, a stethoscope, a spiral notebook with a pen, a pair of glasses, and a smartphone. A large white rounded rectangle with a red border is centered over the laptop.

HAZARDOUS PRODUCTS

HAZARDOUS DRUGS

- Drugs are considered hazardous when they possess ≥ 1 of the following characteristics:
 - Carcinogenicity
 - Teratogenicity or other developmental toxicity
 - Reproductive toxicity
 - Serious organ toxicity at low doses
 - Genotoxicity
- Chemical structures and toxicity profiles of new drugs that mimic existing drugs are determined hazardous based on the above criteria

Requires safe handling precautions!

TWO MAIN CLASSIFICATIONS OF HAZARDOUS DRUGS

Cytotoxic hazardous drugs

- Detrimental/destructive to cells in the body (e.g. cytotoxic, mutagenic, genotoxic, or carcinogenic)
- Such agents are commonly used for cancer treatments
- Associated with the **highest risk** of harm in the event of occupational exposure

Non-cytotoxic hazardous drugs

- Medications (other than cytotoxic) that adversely affect the reproductive system (e.g. teratogenicity, impaired fertility), endocrine system, immune system, respiratory system, or can transmit infection
- Associated with a **possible risk** of harm in the event of occupational exposure

HANDLING HAZARDOUS DRUGS

3 types of hazardous drugs:

- Injectables
- Compromised dosage form (CDF): e.g. counting, splitting, crushing, removing from manufacturer packaging
- Intact oral dosage forms (IODF) tablets/capsules: supplied as a “unit dose” packaging which does not require further manipulation

	Cytotoxic hazardous drugs	Non-cytotoxic hazardous drugs
Preparation area	• Unpackaging shipped drugs requires gloves and chemotherapy gowns • All types require a designated cytotoxic packaging area • Injectables and CDF require class II biological safety cabinet	• Injectables requires regular clean room procedures • CDF: requires hazardous packaging area, class II biological safety cabinet • IODF: designated hazardous packaging area
Protective wear	• Injectables or CDF: wear double gloves, chemotherapy gown, bonnet, and surgical mask • IODF: single nitrile gloves, chemotherapy gown	• Injectables: PPE for IV prep (isolation gown, gloves, surgical mask) • CDF: double gloves, chemotherapy gown, bonnet, surgical mask • IODF: single nitrile gloves
Labelling	• Labelling for use in patient area: for injectables and CDFs: 	• Labelling for use in patient area: for injectables and CDFs: 
Packaging format	• Packaging for transport to patient care area: closed leak-proof plastic bags and identified with symbol (above)	• Packaging for transport to patient care area: closed leak-proof plastic bags and identified with symbol (above)

EXAMPLES OF HAZARDOUS MEDICATIONS*

Standard Hazardous Risk

- Carbamazepine
- Chloramphenicol
- Cidofovir
- Cyclosporine
- Entecavir
- Estrogen/progesterone /combinations of both
- Ganciclovir
- Leflunomide
- Lenalidomide
- Letrozole
- Levonorgestrel
- Medroxyprogesterone
- Methimazole
- Mycophenolate mofetil
- Oxcarbazepine
- Phenytoin
- Propylthiouracil
- Raloxifene
- Rasagiline
- Spironolactone
- Tacrolimus
- Thalidomide

Cytotoxic (C)

- Azathioprine
- Bleomycin
- Cisplatin
- Cyclophosphamide
- Docetaxel
- Doxorubicin
- Fluorouracil
- Gemcitabine
- Hydroxyurea
- Idarubicin
- Mercaptopurine
- Methotrexate

Reproductive Risk (R)

- Anastrozole
- Avapritinib
- Clobazam
- Clonazepam
- Colchicine
- Dihydroergotamine
- Divalproex
- Dronedarone
- Dutasteride
- Finasteride
- Fluconazole
- Gonadotropin chorionic
- Isotretinoin
- Misoprostol
- Paroxetine
- Posaconazole
- Siponimod
- Temazepam
- Testosterone
- Topiramate
- Trastuzumab
- Tretinoin
- Ulipristal
- Valproate/valproic acid
- Vigabatrin
- Voriconazole
- Warfarin
- Ziprasidone
- Zoledronic Acid

* Not exhaustive lists

HAZARDOUS WASTE MANAGEMENT

- Hazardous products must be destroyed in accordance with regulations governing these products
- Pharmaceutical products that are expired or otherwise no longer usable are considered pharmaceutical waste
- The pharmacist or pharmacy technician must ensure the following actions:
 - Medications and sharp or pointed instruments are to be disposed of safely, **in compliance with environmental protection laws** in force in the jurisdiction
 - Medications that will be destroyed must be **safely stored in a location separate from other medications in the inventory**
 - A procedure must be implemented for the destruction of pharmaceutical waste

HAZARDOUS WASTE MANAGEMENT

- Policies and procedures must be developed and followed, complying with local, provincial/territorial and federal requirements
 - All personnel involved must receive the appropriate training on destruction processes **to ensure their own protection** and **to prevent contamination of the premises or the environment**
 - **All equipment, products, and vials** used in the compounding of hazardous sterile preparations must be discarded in a hazardous waste container
 - Hazardous waste containers must be identified with a **self-adhesive label** marked “Hazardous waste – cytotoxic”. Containers should be filled only to **three-quarters of their capacity** and then sealed
 - **Sharps** containers removed from the containment primary engineering control (C-PEC) must be decontaminated and then discarded into a hazardous waste container and sent for destruction
 - **Non-sharps waste** used in the compounding of hazardous sterile preparations must be placed in a hazardous waste container inside the C-PEC or placed in a sealable plastic bag before removal from the C-PEC, and then discarded in a hazardous waste container
 - **Outer gloves** must be removed inside the C-PEC, then placed in a hazardous waste container or sealable plastic bag
 - **All PPE** must be discarded in a hazardous waste container
 - Bins used for hazardous waste must be incinerated; decontamination by autoclave and subsequent burial is prohibited

A top-down view of a medical professional's workspace. A silver stethoscope lies on a light blue surface. To its right is a silver laptop displaying a medical website. Below the stethoscope is a spiral-bound notebook with a black pen resting on it. To the right of the laptop is a pair of glasses and a black smartphone. In the bottom left corner is a white coffee cup with a lid. A large, semi-transparent white box with a red border is centered over the image, containing the text 'PRODUCT QUALITY ASSURANCE'. A diagonal watermark 'Licensed to Shreyash Salunkhe' is visible across the image.

PRODUCT QUALITY ASSURANCE

STORAGE

- All commercial products used for preparations must be **stored immediately** upon receipt in a manner that prevents cross-contamination and incompatibilities
- Product storage specified by the manufacturer must be strictly observed, regardless of where products are stored (e.g. warehouse, pharmacy, delivery vehicle)
- For final compounded preparations or products used for preparations, the **storage temperature must be controlled** and remain within the limits specified
- Use the **general maintenance log** to record information on monitoring the room, refrigerator, and other related controls
- A **biomedical refrigerator** or **freezer** must be available for storing products, ingredients and final compounded sterile preparations that need to be refrigerated or frozen
- Use **alternative storage** if temperature variance occurs or if refrigerators/freezers are being cleaned
- Products that have been stored must be **inspected before use for evidence of deterioration**
- **Identify unusable compounded preparations** using the BUD and expiration dates of commercial products, and then discard

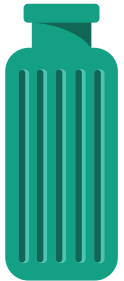
PRODUCT STORAGE CONDITIONS



Some ingredients can degrade when exposed to light



Use amber-coloured bottles to protect from light



Ribbed or fluted bottles are used for products that require external application



Storage conditions must be presented on the product along with the expiry date

SHELF LIFE

- Determines how long a product can be on the shelves in the dispensary
- Medications have different physical, chemical and bacterial properties which affect deterioration rates
- Stability of the ingredients is dependent on the admixture and on how the product was created

THE “BEYOND-USE DATE”

- The **Beyond-Use Date (BUD)** refers to the date after which a non-sterile preparation should not be used
- The criteria for assessing the BUD is **different from manufactured drug products expiry dates**
- BUD is dependent on literature and documentation on the specific stability of the active ingredient

Parameters to consider when assigning BUD

- | | |
|--------------------------------------|---|
| • Nature of ingredient(s) to be used | • Compounding method |
| • Degradation mechanisms | • Compatibility |
| • Dosage forms | • Potential for microbial proliferation |
| • Container to be packaged in | • Expected storage conditions |
| • Intended use & duration of therapy | |

THE BUD OF NON-STERILE PREPARATIONS

Non-aqueous formulations

- The BUD does not surpass the earliest expiry date of any of the active ingredients or 6 months, whichever is earlier

Water-containing oral formulations

- Stored at controlled cold temperatures, BUD cannot exceed more than 14 days

Water-containing dermal/topical, mucosal liquid and semi-solid formulations

- The BUD cannot exceed more than 30 days

THE BUD OF COMMERCIALY AVAILABLE PRODUCTS

Single-dose vial

- If the vial is punctured, the BUD is 6 hours, after which the vial can no longer be used
- If removed from the hood, discard immediately



Affix a label to the vial indicating the time of needle puncture

Open ampoule

- No storage is permitted, therefore, no BUD applies

Multiple-dose container

- Usually contains a preservative
- The BUD is 28 days unless otherwise specified by the manufacturer or visible contamination is present

THE BUD OF STERILE PREPARATIONS

- A sterile unit is a vial, ampoule or bag of drug, or diluent
 - E.g. 1 vial of water for injection represents 1 “sterile unit”
- Contamination risk levels
 - **Low**: drug prepared for one patient, simple aseptic transfer technique, final product compounded using up to 3 “sterile units”
 - **Medium**: batch preparations, complex manipulations, prolonged preparation time, final product compounded using ≥ 4 “sterile units”
 - **High**: non-sterile ingredients used before terminal sterilization, improper garbing by personnel

	BUD without sterility testing		
Risk of contamination	At controlled room temperature	With storage in refrigerator	With storage in freezer
Low	48 hours	14 days	45 days
Medium	30 hours	9 days	45 days
High	24 hours	3 days	45 days

LABELLING REQUIREMENTS

MAIN AND SUPPLEMENTARY LABELS

Each container for a compounded preparation must be labelled

- **Pharmacy identification (name, address, telephone number of the compounder's or dispenser's pharmacy)**
- **Drug identification (APIs, source, concentration, form, route of administration, volume, solute, amount prepared)**
- **Overfill volume (sterile preparations)**
- **Special precautions and storage conditions (auxiliary labels)**
- **Date when the sterile preparation was compounded**
- **BUD**
- **Preparation batch number**
- **All information required by provincial/territorial legislation and regulations**



The compounding pharmacist/pharmacy technician must similarly label sterile preparations that have been compounded at the request of another pharmacy

At the dispensing pharmacy, another label is added containing all the information required by the provincial/territorial regulatory authority. Both labels are retained on the preparation(s)

PACKAGING REQUIREMENTS

- Must maintain the preparation's stability, integrity and storage conditions
- Indicate storage conditions on the final package (e.g. protect from light, temperature)
- Indicate additional precautions on the final packaging (e.g. if the product is an irritant)
- Indicate transport precautions (e.g. temperature, fragility, safety) and instructions (name and address of patient) on the outside packaging of each item

The packaging procedure must specify:

- Equipment used to prevent breakage, contamination, spills or degradation during transport
- Method used to confirm whether the temperature of the compounded preparation(s) has been maintained during transport
 - E.g. certified cooler, temperature maintenance indicator, min./max. thermometer
- Packaging used to protect against extreme temperatures (freezing and excessive heat) unless evidence exists demonstrating the product's stability at these temperatures
 - E.g. maintain refrigerated compounds at 2-8°C; for compounds kept at room temperature, maintain at 19-25°C

QUALITY ASSURANCE PROGRAM

	Example	Frequency of review
Facilities	Compounding areas must be clean, orderly, well-maintained, and suitable for risk level	Every 6 months, at installation, after repairs, or if contamination is suspected
	Daily temperature and humidity logs	Monthly
Equipment	Containment primary engineering control (C-PEC) must be certified	Before first use, every 6 months, after installation/repairs, or if contamination is suspected
	Temperature probes must be calibrated	Calibrate annually, verify logs monthly
	Instruments must be checked for operational performance	Monthly
Personnel	Skills and competency assessments	At initial qualification, and then periodically
Final Preparation	Master Formulation Records (MFRs)	Annually or when new data emerges
	Product appearance (clarity, odour, colour, consistency, pH)	Verify against every prescription
Documentation	Policies and procedures	Every 3 years or sooner
	Compounding records and logs	Quarterly to ensure regulatory compliance
	References and Safety Data Sheets (SDS)	Annually

TAKEAWAY

Compounding is the practice of making a pharmaceutical product to satisfy a therapeutic need as there is no commercially available alternative

NAPRA has Model Standards for non-sterile and sterile compounding (non-hazardous and hazardous preparations)

Pharmacy professionals must apply appropriate policies and procedures (e.g. hand hygiene, PPE, airflow principle, facilities, personnel behaviour, documentation, labelling, storage, distribution to patients, quality controls, maintenance)

Various compounding techniques (trituration, levigation, geometric dilution) are used to ensure preparations are accurate, homogenous, and stable

Hazardous drugs may be cytotoxic or non-cytotoxic, requiring different handling procedures

CASE STUDY

AB is a 64-year-old male with metastatic prostate cancer. His oncologist has prescribed an oral suspension of cyclosporine 35 mg/mL for off-label use to manage graft-versus-host-like symptoms after a stem cell transplant. AB has difficulty swallowing capsules due to mucositis from his chemotherapy regimen. The commercial oral solution is currently unavailable, so the prescriber requests a compounded suspension. The prescription reads:

Cyclosporine 35 mg/mL oral suspension

Sig: 5 mL PO BID every other day

Quantity: 120 mL

Your compounding pharmacy will use cyclosporine capsules, microcrystalline cellulose as a suspending agent, a sucrose-based flavoured aqueous syrup, and purified water as the diluent to prepare this prescription.

CASE STUDY - QUESTIONS

1. Which of the following is the most appropriate engineering control for compounding this preparation?
 - a) Class I biological safety cabinet (BSC)
 - b) Class II biological safety cabinet (BSC)
 - c) Laminar flow hood
 - d) No hood required, standard compounding area is acceptable
2. What personal protective equipment (PPE) is most appropriate for compounding this preparation?
 - a) Gloves only
 - b) Gloves and a surgical mask
 - c) Double chemotherapy gloves, gown, and eye protection
 - d) Double gloves, N95 respirator mask, and sterile gown
3. Which compounding technique is most appropriate to ensure uniform particle dispersion in this suspension?
 - a) Dissolve cyclosporine in water before adding remaining ingredients
 - b) Triturate the drug into a fine powder, levigate with a suspending vehicle, add sufficient quantity for the final volume
 - c) Triturate the drug into a fine powder, add directly into flavoured syrup to make the final volume, shake vigorously
 - d) Mix cyclosporine with isopropyl alcohol before adding remaining ingredients
4. What is the most appropriate beyond-use-date (BUD) for this suspension?
 - a) 14 days refrigerated
 - b) 30 days refrigerated
 - c) 90 days refrigerated
 - d) 180 days refrigerated

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CHANGE LOG

May 2025

- Slide 9: Changed BUD information based on NAPRA guidance

July 2025

- Formatting, grammar, and spelling changes throughout
- Slide 3: Updated legend
- Slide 4: Added footnote
- Slide 23: Added cyclosporine to standard hazardous drugs list, moved anastrozole to reproductive risk, removed diethylstilbesterol and zonisamide
- Slide 24: Changed slide title to "Takeaways"
- Slides 25, 26: Added case study and questions
- Slide 27: Updated references

September 2025

- Formatting, grammar, and spelling changes throughout
- Significant changes made throughout, rearranged slide order
- Removed slide on hospital compounding (duplicate information)
- Slide 3: Updated legend
- Slide 5: Expanded information
- Slides 6-8, 10, 18-23: New slides added
- Slides 9, 15, 24, 31: Separated slides with subtitle slides
- Slide 16: Replaced "Hospital" with "Sterile" in slide title
- Slide 17: Changed slide title, added note on containment primary engineering controls and changed table subheadings
- Slide 28: Updated lists
- Slides 29, 30: Added hazardous waste management slides
- Slide 32: New slide added on packaging
- Slide 36: Changed slide title to include 'non-sterile preparations'
- Slide 37: New slide added to include the BUD of commercial products
- Slide 38: New slide added to include the BUD of sterile preparations
- Slide 39: Added information
- Slide 40: Added details to packaging requirements
- Slide 41: New slide added on quality assurance program components
- Slide 42: Updated Takeaway
- Slide 45: Updated References

March 2026

- Slide 2: Updated NAPRA competencies
- Minor formatting edits made